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Schrijf in poolcoördinaten:

$$x^2 + y^2 = 2x(3y^2 - x^2)$$

$$\Rightarrow (r \cos \theta)^2 + (r \sin \theta)^2 = 2r \cos \theta \cdot (3(r \sin \theta)^2 - (r \cos \theta)^2)$$

$$\Leftrightarrow r^2 = 6r^3 \sin^2 \theta \cos \theta - 2r^3 \cos^3 \theta$$

$$\Leftrightarrow \frac{1}{r} = 6 \sin^2 \theta \cos \theta - 2 \cos^3 \theta \text{ of } r = 0$$

$$\Leftrightarrow r = \frac{1}{6 \sin^2 \theta \cos \theta - 2 \cos^3 \theta} \text{ of } r = 0$$

References